

Urban Tree Methods with the USDA Urban Tree Database

Rebecca Bronstein

2026-04-01

Contents

0.1	Data Source	1
0.2	Why Tree Methods Fit This Dataset	1
0.3	Data Snapshot	1
0.4	Classification Tree: Street Trees vs. Park-Like Trees	2
0.5	Regression Tree: What Predicts Tree Height?	4
0.6	Rule-Based Tree: Which Trees Deliver High Car Shade?	5
0.7	Bagged Trees: Does the Site Classifier Become More Stable?	7
0.8	Conclusion	8

0.1 Data Source

This report uses the USDA Forest Service Research Data Archive publication **Urban tree database** and, specifically, the local file `Data/TS3_Raw_tree_data.csv` included in this folder. The archive describes the publication as a multi-city urban tree dataset covering more than 14,000 measured street and park trees across 17 cities in the United States. Source page: <https://www.fs.usda.gov/rds/archive/catalog/RDS-2016-0005>.

0.2 Why Tree Methods Fit This Dataset

This dataset mixes ecology, urban form, and management questions in one table. Tree methods work especially well here because they can combine species categories, regional labels, and structural measurements without forcing everything into a single linear relationship.

In this notebook I use tree methods to answer four practical questions:

1. Can tree structure and context tell us whether a record comes from a street tree or a more park-like setting?
2. Which features matter most when predicting tree height?
3. Can a rule-based model explain which trees provide high car shade?
4. Does bagging improve the street-versus-park-like classification?

0.3 Data Snapshot

```
kable(overview, caption = "Summary of the USDA urban tree dataset used in this  
↪ report.")
```

Table 1: Summary of the USDA urban tree dataset used in this report.

Measure	Value
Total tree records	14487
Distinct cities	17
Distinct climate regions	17
Street trees	8987
Non-street trees grouped as park-like	5500
Distinct tree types	12

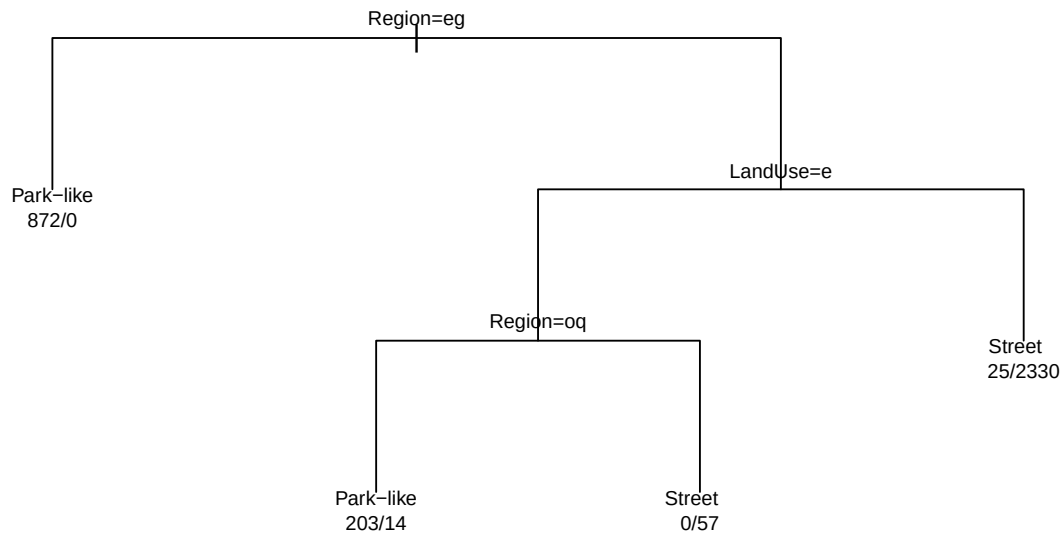
0.4 Classification Tree: Street Trees vs. Park-Like Trees

The original Park/Street field has a few small categories such as nursery and regional big tree. To make the result easier to explain, I collapse the outcome into two classes:

1. Street
2. Park-like

This lets the tree model act like a site classifier: based on size, canopy form, and land-use context, what kind of planting environment does a tree look like it belongs to?

```
site_tree <- rpart(Site.Type ~ ., data = train_site, method = "class")  
plot(site_tree, uniform = TRUE, margin = 0.08)  
text(site_tree, use.n = TRUE, cex = 0.7)
```



```

site_pred <- predict(site_tree, test_site, type = "class")
site_metrics <- classification_metrics(test_site$Site.Type, site_pred)
site_importance <- top_importance(site_tree)

kable(site_metrics, caption = "Classification performance for predicting site type.")

```

Table 2: Classification performance for predicting site type.

Accuracy	Kappa
0.987	0.971

```

kable(site_importance, caption = "Most influential variables in the site-type tree.")

```

Table 3: Most influential variables in the site-type tree.

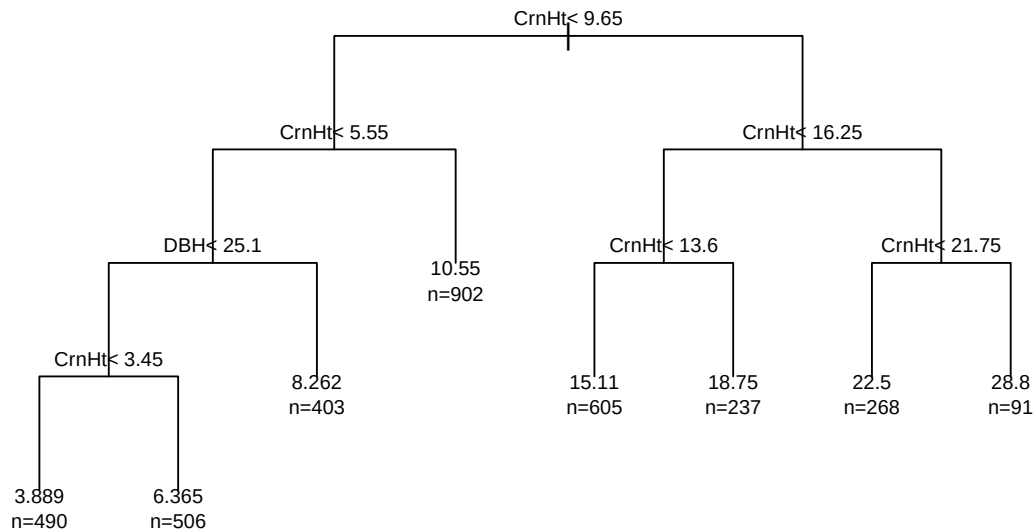
Variable	Importance
Region	1171.33
LandUse	645.09
Leaf	26.31
WireConf	24.95
Setback	15.25
Age	15.03

This is a useful planning interpretation of the dataset. The strongest variables are region, land use, and setback, which suggests that site context matters at least as much as raw tree dimensions when distinguishing street trees from trees in park-like settings.

0.5 Regression Tree: What Predicts Tree Height?

Urban tree height is shaped by both biology and management. A regression tree is useful here because it can split on crown height, diameter, and leaf area in a way that feels close to how an arborist might reason through the problem.

```
height_tree <- rpart(TreeHt ~ ., data = train_height, method = "anova")
plot(height_tree, uniform = TRUE, margin = 0.08)
text(height_tree, use.n = TRUE, cex = 0.7)
```



```
height_pred <- predict(height_tree, test_height)
height_metrics <- regression_metrics(test_height$TreeHt, height_pred)
height_importance <- top_importance(height_tree)
```

```
kable(height_metrics, caption = "Regression performance for predicting tree height in  
↪ meters.")
```

Table 4: Regression performance for predicting tree height in meters.

RMSE	MAE
2.22	1.62

```
kable(height_importance, caption = "Most influential variables in the tree-height
↪ model.")
```

Table 5: Most influential variables in the tree-height model.

Variable	Importance
CrnHt	124011.20
Leaf	65971.88
AvgCdia	56974.27
DBH	50560.95
Age	32469.10
TreeType	15566.46

The importance scores usually put crown height, leaf area, average crown diameter, and DBH near the top. That makes the model easy to justify: urban tree height is strongly tied to overall crown architecture, not just one diameter measurement.

0.6 Rule-Based Tree: Which Trees Deliver High Car Shade?

Shade is one of the clearest urban ecosystem services in this dataset. To make that idea more concrete, I create a binary target called Big.Shade, where trees with higher CarShade values are labeled High shade.

The rule model asks a planning-style question: which combinations of region and tree structure tend to produce stronger shade benefits?

```
shade_rules <- C5.0(Big.Shade ~ ., data = train_shade, rules = TRUE)
shade_pred <- predict(shade_rules, test_shade)
shade_metrics <- classification_metrics(test_shade$Big.Shade, shade_pred)
shade_usage <- as.data.frame(C5imp(shade_rules, metric = "usage")) %>%
  rownames_to_column(var = "Variable") %>%
  rename(Usage = Overall) %>%
  arrange(desc(Usage)) %>%
  mutate(Usage = round(Usage, 2))

kable(shade_metrics, caption = "Rule-based performance for predicting high car shade.")
```

Table 6: Rule-based performance for predicting high car shade.

Accuracy	Kappa
0.935	0.703

```
kable(head(shade_usage, 6), caption = "Most-used predictors in the shade rules.")
```

Table 7: Most-used predictors in the shade rules.

Variable	Usage
Region	100
Park.Street	0
TreeType	0
DBH	0
TreeHt	0
AvgCdia	0

```
summary(shade_rules)
```

```
##
## Call:
## C5.0.formula(formula = Big.Shade ~ ., data = train_shade, rules = TRUE)
##
##
## C5.0 [Release 2.07 GPL Edition]      Wed Apr  1 21:36:56 2026
## -----
##
## Class specified by attribute `outcome'
##
## Read 3501 cases (8 attributes) from undefined.data
##
## Rules:
##
## Rule 1: (400/77, lift 6.0)
##   Region in {NMtnPr, Piedmt}
##   ->  class High shade  [0.806]
##
## Rule 2: (3101/150, lift 1.1)
##   Region in {CenFla, GulfCo, InlEmp, InlVal, InterW, LoMidW, MidWst,
##             NoCalC, NoEast, PacfNW, SacVal, SoCalC, SWDsrt, TpIntW,
##             Tropic}
##   ->  class Low shade  [0.951]
##
## Default class: Low shade
##
##
## Evaluation on training data (3501 cases):
##
##           Rules
##   -----
##   No      Errors
##
##       2  227( 6.5%)  <<
```

```
##      (a)   (b)   <-classified as
##      ----  ----
##      323   150   (a): class High shade
##      77   2951   (b): class Low shade
##
##
## Attribute usage:
##
## 100.00% Region
##
##
## Time: 0.0 secs
```

This section is intentionally more interpretive than predictive. Even when the rules are short, they help translate the dataset into a management idea: some regional and structural profiles are much more likely to correspond to trees that deliver stronger shade.

0.7 Bagged Trees: Does the Site Classifier Become More Stable?

A single classification tree is easy to explain, but it can change if the sample shifts. Bagging gives us a way to ask whether the street-versus-park-like pattern is robust or whether it depends too heavily on one tree.

```
bag_model <- bagging(Site.Type ~ ., data = train_site, nbagg = 25)
bag_pred <- predict(bag_model, test_site)

if (is.list(bag_pred) && "class" %in% names(bag_pred)) {
  bag_pred <- bag_pred$class
}

bag_pred <- factor(bag_pred, levels = levels(test_site$Site.Type))

bag_metrics <- classification_metrics(test_site$Site.Type, bag_pred) %>%
  mutate(Model = "Bagged trees")

single_tree_metrics <- site_metrics %>%
  mutate(Model = "Single tree")

comparison <- bind_rows(single_tree_metrics, bag_metrics) %>%
  select(Model, Accuracy, Kappa)

kable(comparison, caption = "Comparing a single site-type tree with a bagged
  ↪ ensemble.")
```

Table 8: Comparing a single site-type tree with a bagged ensemble.

Model	Accuracy	Kappa
Single tree	0.987	0.971
Bagged trees	0.996	0.991

If the bagged model improves only a little, that is still informative. It means the site signal in this USDA dataset is already strong enough that a single tree captures much of the story, while the ensemble mostly adds stability.

0.8 Conclusion

This USDA urban tree dataset is especially well suited to tree-based methods because it combines species, structure, region, and land-use information in one place.

The models here suggest four useful takeaways:

1. Site context can be inferred very accurately from a tree's regional and structural profile.
2. Tree height behaves like a crown-architecture problem as much as a diameter problem.
3. Rule-based models make ecosystem-service questions, like shade, easier to explain in plain language.
4. Bagging confirms that the street-versus-park-like distinction is a stable pattern, not a fragile one.

That makes this archive a strong example of how tree methods can support both ecological interpretation and urban forestry planning.